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Development and Implementation of Coiled Tubing Technology for Gas Lift Systems in Artificial Lift

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Abstract

This paper presents the development and successful field implementation of coiled tubing technology for enhanced gas lift systems in artificial lift operations. The study focuses on optimizing lift efficiency through tailored string designs, precise gas injection placement, and improved operational reliability, targeting enhanced production from mature and underperforming wells while reducing intervention frequency and operational costs.

The methodology encompasses comprehensive engineering design of coiled tubing systems specifically tailored for gas lift operations, including optimal tubing material selection, strategic injection depth determination, and gas lift system configuration based on individual well characteristics. Field implementation involved detailed planning, real-time monitoring, and optimization of gas injection rates, with performance evaluation through continuous data analysis to ensure efficient production rates and system reliability. Post-deployment assessments were conducted to refine the technology for future applications, minimizing operational downtime and enhancing overall well performance.

Implementation of coiled tubing for gas lift systems delivered significant improvements in well productivity and operational efficiency. Key achievements include enhanced gas lift efficiency, superior injection rate control, and improved adaptability to varying well conditions. Optimized tubing configurations and gas injection strategies contributed to more stable production, particularly in mature wells, while real-time system monitoring enabled precise adjustments and reduced intervention frequency. The technology demonstrated faster deployment capabilities and greater flexibility to respond to changing reservoir conditions compared to traditional methods, resulting in reduced operational downtime and intervention costs.

The study concludes that coiled tubing technology provides an effective and reliable solution for enhancing gas lift performance in artificial lift operations. Continuous monitoring and adaptation to specific well conditions are critical to maximizing technology benefits. Future applications should focus on integrating advanced diagnostic tools for real-time optimization and further reducing costly well interventions. The success of this method validates its potential as a cost-effective solution for gas lift

operations in mature fields, offering engineers actionable insights to optimize operations and extend well lifespan through innovative design methodologies and performance evaluation metrics.

Introduction

This case study was conducted in a field located 680 kilometers from Bogotá, Colombia. Access roads are limited and challenging due to incomplete pavement and considerable distance from main transportation routes, making resource mobilization to the wellsite extremely difficult. The field produces approximately 2,000 barrels per day of fluid at 500 psi.

The optimization of artificial lift systems remains a persistent challenge in the oil and gas industry, particularly in mature fields and underperforming wells. In this context, coiled tubing technology has emerged as an innovative solution to enhance gas lift system efficiency. Its ability to enable rapid, precise, and minimally invasive interventions has opened new possibilities for gas lift application design and operation.

This paper presents the development and field implementation of an advanced methodology that integrates coiled tubing into gas lift operations, aiming to maximize well productivity and reduce operational costs. The study encompasses system engineering design, candidate well selection, deployment strategy, and performance evaluation under real operating conditions. Special emphasis is placed on customized tubing designs, precise gas injection placement, and real-time monitoring for operational adjustments.

The results demonstrate significant improvements in gas lift efficiency, production stability, and reduction in corrective interventions. This implementation not only validates coiled tubing's potential as a versatile and reliable tool but also provides a solid foundation for future applications incorporating advanced diagnostic tools and continuous optimization strategies.

Statement of Theory and Definitions

The application of coiled tubing in gas lift systems is grounded in artificial lift optimization principles, which aim to enhance hydrocarbon recovery by improving fluid flow dynamics and minimizing operational inefficiencies. Gas lift, as an artificial lift method, involves injecting high-pressure gas into the production tubing to reduce the hydrostatic pressure of the fluid column, thereby facilitating upward movement of reservoir fluids. Traditional gas lift systems often rely on fixed injection points and conventional tubing configurations, which may limit adaptability and efficiency under dynamic well conditions.

Coiled tubing refers to a continuous length of small-diameter steel pipe wound on a spool, capable of being inserted into wells without requiring a conventional rig. Its flexibility and deployability make it ideal for live well interventions, allowing precise placement of gas injection points and real-time system adjustments. In gas lift operations, coiled tubing enables tailored string designs and optimized injection depths, which are critical for enhancing lift performance and reducing well intervention frequency.

Coiled Tubing Gas Lift Installation

The operation begins with verification of the final well schematic and mechanical conditions to ensure displacement calculation accuracy. The bottom hole assembly (BHA) was composed of two different types of check valves, which provided a dual barrier to prevent inflow into the CT, and a customized pump-out plug that allowed gas injection and pressure testing of the system once installed. Prior to running in hole (RIH) with the coiled tubing, wellhead pressures are equalized, and the swab/master valve is opened, recording the number of turns. Initial descent is performed at low speed (10 ft/min), gradually increasing based on pipe weight behavior and well conditions.

During RIH, pull tests are conducted every 2,000–3,000 ft, and speed is reduced when passing through internal restrictions. Upon reaching target depth (~9,500 ft), the coiled tubing is placed under tension, BOP

slips and rams are activated, and pressure is bled off from the control system. The coiled tubing is cut 6 ft above the BOP, followed by installation of the external connector and inner hanger, which are tested under load and pressure.

The installation bottom hole assembly (BHA) is assembled according to established design specifications, including tools such as the GS pulling tool, which undergoes hydraulic testing. The pulling tool is manually connected to the inner hanger, engagement is verified, and the stack is adjusted. The system is pressurized, and final coiled tubing installation against the BOP is completed, ensuring proper weight transfer.

Subsequently, the CT is set in the hanger, and the pulling tool is released via controlled fluid circulation, seal integrity is tested, and the Pump Out Plug is activated through staged pressurization. Nitrogen (N₂) pumping is performed for injection testing, with parameters monitored according to customer specifications.

Finally, hanger sealing is verified, accessories are disconnected, production lines are installed, and complete rig down of coiled tubing equipment is carried out.

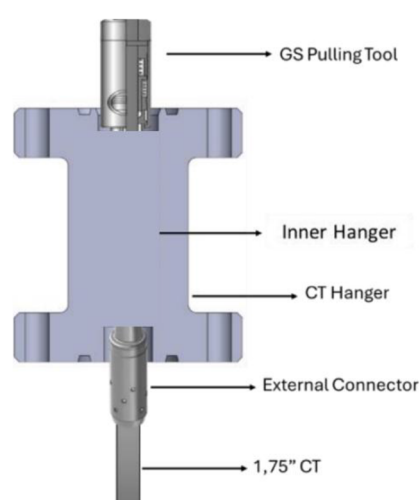


Figure 1—GS Pulling Tool and Inner Hanger Surface Assembly

When to Deploy Coiled Tubing in Gas Lift?

Coiled Tubing Gas Lift is typically deployed in oil and gas wells under the following operational and reservoir conditions:

1. Low Reservoir Pressure: Enables deep gas injection to enhance artificial lift when natural energy is insufficient.
2. Restricted or Damaged Tubing: CT provides a bypass for gas injection without removing existing production tubing.
3. Temporary or Emergency Lift: Suitable for short-term production or during interventions; fast deployment without a workover rig.
4. High Water Cut / Liquid Loading: Reduces hydrostatic pressure and unloads liquids, improving gas flow.
5. Need for Deep Gas Injection: Allows gas injection closer to the producing zone, beyond the reach of conventional valves.
6. Complex or Deviated Well Geometry: CT's flexibility enables gas lift in horizontal or highly deviated wells where conventional methods are limited.

Parameters to consider when performing a Coiled Tubing Gas Lift

For successful Coiled Tubing Gas Lift installation, the following parameters must be considered:

Design and Planning Parameters

- Final well schematic: Confirm trajectories, depths, internal restrictions, and mechanical conditions
- Displacement calculations: Validate fluid volumes, pressure profiles, and expected flow behavior
- Bottom Hole Assembly (BHA) design: Selection of tools including GS pulling tool, inner hanger, valves, connectors, and other components
- Coiled tubing material and diameter: Based on well conditions and gas injection requirements
- Target installation depth: Confirmed on-site with customer representative

Operational Parameters

- Run-In-Hole (RIH) speed:
 - Initial: 10 ft/min (first 50 ft)
 - Normal: 45–50 ft/min (depending on pipe weight behavior)
 - Reduced speed in critical zones: 10 ft/min at ID changes or restrictions
- Pull tests: Every 2,000–3,000 ft or as required by well conditions
- String weight monitoring: Continuous tracking to avoid overload or tension loss
- Wellhead pressure: Equalization prior to valve opening
- BOP and connector pressure testing:
 - BOP: According to pressure test matrix
 - Connectors: Up to 7,000 psi
- Tool release pumping pressure: 1.0 bpm flow rate with staged pressure up to 4,000 psi
- Pump Out Plug shear pressure: Continue pressurization until bronze shear pins rupture
- Nitrogen (N₂) flow rate: Start at 350 scfm, with maximum of approximately 4 MMSCFD

Force Calculations

Weight simulations consider the well's completion and characteristics, including the survey, which in this well type reaches up to 92°. This weight analysis and interpretation during Run In Hole (RIH) and Pull Out Of Hole (POOH) of the CT are important to verify whether the CT can reach target depths.

The weight simulations shown in [Fig. 2](#) show four curves with the following descriptions:

- Lock Up: A weight indicator reading below this indicates that no additional force transmitted from surface will reach the downhole end of the CT at the BHA end. Below this value, any additional surface weight will not reach the CT end, resulting in no further CT advancement along the wellbore.
- Yield: The maximum surface weight that can be applied to the CT.
- Trip In: The expected weight indicator values during RIH. These values must always be greater than the lockup curve to realistically predict reaching the bottom of the well without any lockup issues.
- Trip Out: Expected weight indicator values during POOH from bottom hole back to surface. These values should always remain below the Yield curve to conduct operations without overstressing the CT.

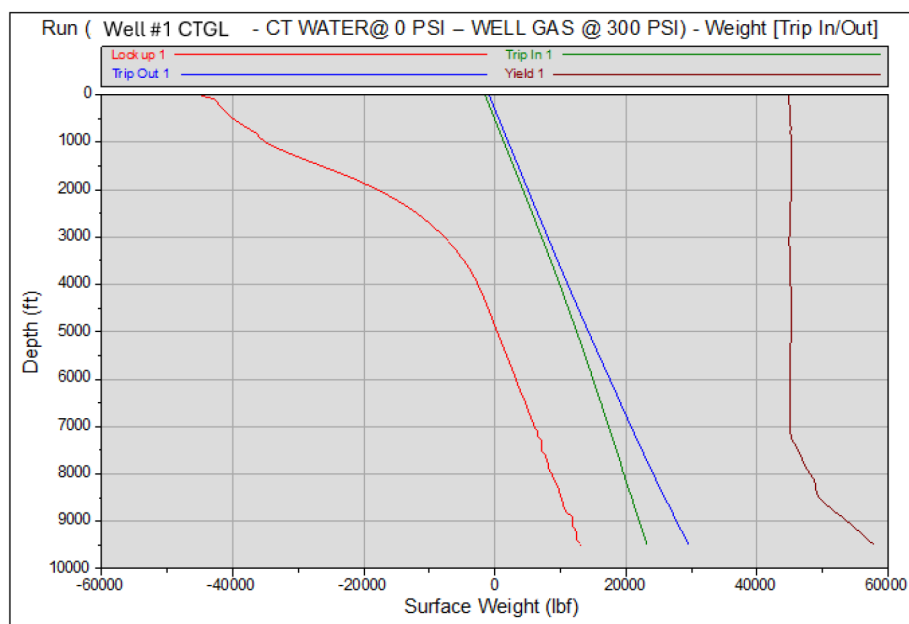


Figure 2—Weight considerations.

Rig-Up and Surface Testing

1. Conduct pre-operational meeting with all personnel involved in the operation
2. Install external dimple connector for Coiled Tubing and perform pull test at 30,000 lbf and pressure test at 7,000 psi
3. Calibrate Coiled Tubing using drift ball
4. Connect Pump Out Plug to connector and perform surface activation test using 3 copper shear pins rated at 1,100 psi each $\pm 15\%$, recording actual shear pressure of each pin
5. Once Pump Out Plug is activated, define number of pins to be installed in coordination with client and redress tool accordingly. Well fluid level must be considered for this step.

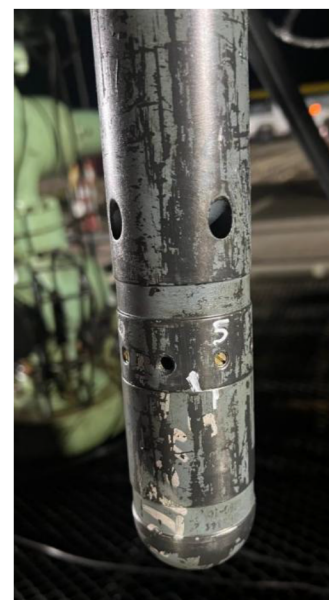
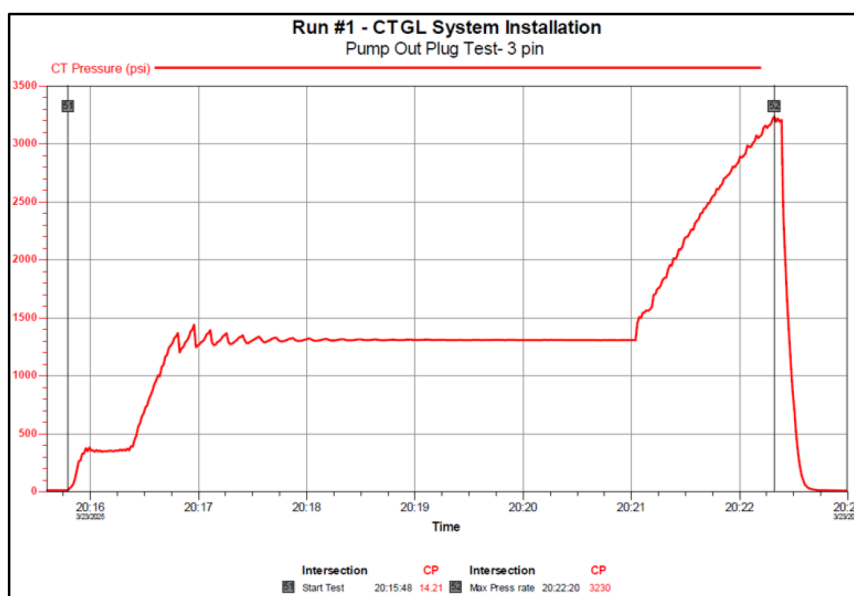


Figure 3—Pump Out Plug Activation.

Purpose in Gas Lift Installations with Coiled Tubing

In Coiled Tubing (CT) Gas Lift operations, the Pump Out Plug (POP) plays a critical role in ensuring the integrity, safety, and controlled activation of the system during installation and commissioning. Its functions are multifaceted and essential for maintaining operational reliability:

- Allow system pressure testing: Prior to deployment into the wellbore, the POP enables isolation of the CT string and Bottom Hole Assembly (BHA), allowing the entire system to be pressurized from surface. This facilitates comprehensive pressure testing to confirm the integrity of connections, seals, and components, ensuring readiness for downhole operations.
- Controls system activation: The POP is designed to shear or release at a predetermined pressure threshold—typically governed by shear pins (e.g., three copper pins rated at 1,100 psi $\pm$ 15%). This controlled release mechanism ensures that downhole fluid communication only occurs once the system reaches the required activation pressure, preventing premature flow and enabling precise operational timing.
- Prevents premature fluid entry: In wells that are already producing or have significant bottomhole pressure, the POP acts as a barrier to prevent early fluid ingress into the CT string. This is particularly important during installation, as it avoids unintended system activation or interference with the BHA before full deployment and validation.
- Facilitates safe BHA installation: By maintaining isolation within the CT string, the POP allows for safe handling and installation of the BHA. It prevents accidental activation of internal components due to pressure surges or fluid movement, thereby reducing the risk of mechanical failure or operational hazards during run-in-hole procedures.

Coiled Tubing Gas Lift Design Process

- Preparation: Equipment and tool verification, drift ball compatibility check, and pre-job safety meeting.
- Rig-Up & Initial Testing: Install and test external dimple connector, calibrate CT with drift ball, activate Pump Out Plug with shear pins @ 1,100 psi $\pm$ 15%.
- BHA Installation: Assemble up to the ball check valve, pressure test connections, and position CT at target depth.
- Cutting & Final Connection: Cut CT, install external connector, torque Inner Hanger, perform pull test, and release GS Pulling Tool.
- System Activation: Gradually pressurize up to 4,500 psi, shear Pump Out Plug pins, and initiate gas injection.

Operational Description

A specialized intervention was carried out to install a gas lift system using coiled tubing (CT), aiming to enhance well production efficiency through an optimized artificial lift solution.

Main Activities Performed:

- CT Run-In-Hole (RIH): The CT was deployed from a zero depth of 17.5 ft to a final depth of 9660 ft at controlled speeds (10–50 ft/min), with pull tests conducted every 2000 ft. No circulation was performed during this phase.
- CT Positioning and Cutting: The CT was placed under tension at 9575 ft, marked, and secured by sequentially closing the BOP rams. The CT was then cut, and the end was prepared for installation of the external connector and inner hanger.

- BHA Assemblies Installation: BHA were installed, including connectors, the GS pulling tool, and dummy hangers. Pressure and pull tests were conducted to validate the integrity of all connections.
- GS Pulling Tool Testing: Multiple hydraulic release attempts were made with the GS pulling tool. After mechanical adjustments and redressing, a successful release was achieved under controlled flow and pressure conditions.
- Final CT Positioning and Release: The assemblies were manually connected, additional pull tests were performed, and the CT was released from the BOP rams. The string was set in the surface body hanger, fully transferring the string weight.
- Seal and Injection Testing: The integrity of the hanger seals was verified through staged pressurization up to 3400 psi, successfully activating the Pump Out Plug. Fresh water was circulated through the gas lift system without significant pressure increase.

Final Displacement and Closure: The manual gate valve was closed, and the remaining CT fluid volume was displaced using 253 gallons of nitrogen, completing the intervention.

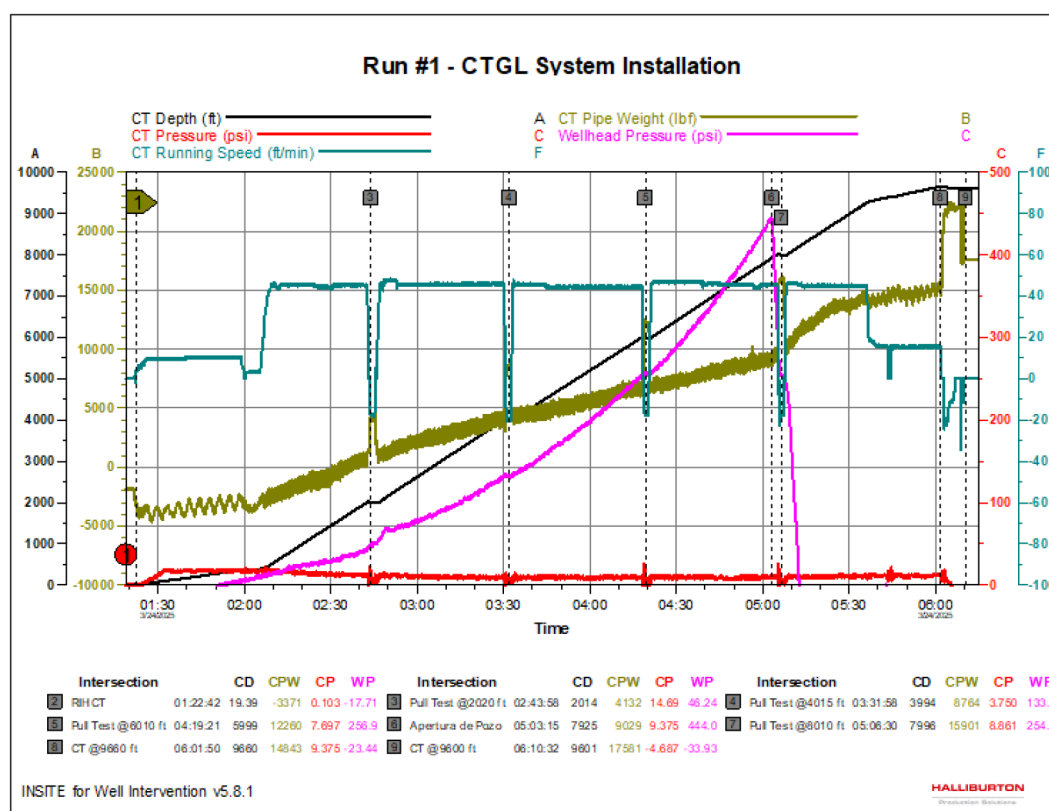


Figure 4—Coiled Tubing Gas Lift Installation.

Upon completion of mechanical installation of the coiled tubing (CT) string and system securing via inner hanger, the Blind/Shear Rams of the Dual BOP were closed. Nitrogen (N₂) injection flow was aligned through the flow cross, utilizing the hung CT string as injection conduit. Return lines were directed to production, with wing line choke set to ½ inch.

Nitrogen injection commenced at 400 scfm to evaluate installed gas lift system performance. During this phase, typical hydrostatic column loading and unloading behavior was observed, with intermittent surface fluid discharges occurring in batches. The initial batch exhibited the highest discharge volume and peak injection pressure, followed by progressive decline in both hydrocarbon output and injection pressure. As

the fluid column was charged and pushed to surface, the system required increasingly longer durations to complete each cycle, indicating reduced displacement efficiency over time.

The system exhibited a pattern of six hydrostatic column charging cycles, characterized by decreasing pressure peaks and extended discharge durations, suggesting system stabilization and diminishing push efficiency as fluid columns depleted.

Recorded Operational Parameters

- N₂ Injection Rate: 400 scfm
- Total N₂ Volume Pumped: 7000 gallons
- CTGL System BHA Depth: 9000 ft
- Wing Choke Setting: ½ inch
- Maximum Injection Pressure: 2560 psi
- Minimum Injection Pressure: 492 psi
- Number of Column Charging Cycles: 6
- Injection Pressure Peaks per Cycle: 2560, 2129, 1727, 1729, 1254, 853 psi
- Minimum Pressure at End of Surface Discharges: 1081, 785, 554, 497, 493 psi

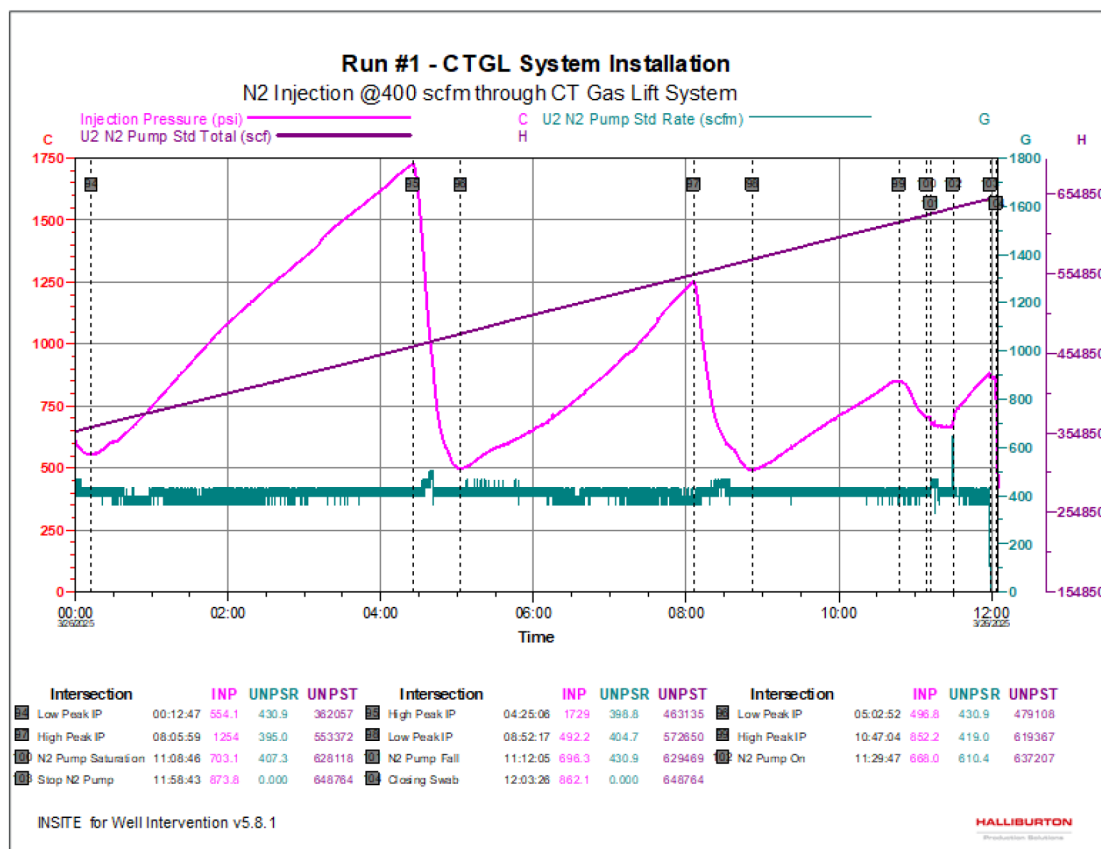


Figure 5—N₂ Injection CT Gas Lift.

Conclusions

- The successful installation and commissioning of the Coiled Tubing (CT) Gas Lift system demonstrated the effectiveness of the planned intervention strategy and the robustness of the operational execution. The deployment adhered strictly to the established work program, achieving all technical and safety objectives without incidents or environmental impact, underscoring the high level of coordination among the involved service providers.
- The intervention involved the installation of a Velocity String-Gas Lift system utilizing 9,000 feet of CT, integrated with a 2.125" Injection Bottom Hole Assembly (BHA). The CT string was securely anchored at surface using a 4.06" 10K CT Hanger assembly, ensuring mechanical integrity and pressure containment.
- Well control was managed through a multi-barrier approach. Primary barriers included a Pump Out Plug, hydrostatic pressure maintained by water-filled CT, a Dual Back Pressure Valve, and a Ball Check Valve. Secondary barriers were provided by a Well Pressure Control Equipment (WPCE) stack comprising two surface-mounted Blowout Preventers (BOPs) rated at 15K and 10K psi, in full compliance with API specifications.
- The operation required the execution of complex mechanical and hydraulic procedures, including tool integrity tests, connector validations, and pressure integrity verifications. All activities were completed without delays or deviations from safety protocols, reflecting meticulous planning and operational discipline.
- Performance evaluation of the installed CT Gas Lift system was conducted via nitrogen injection at 400 scfm. The system exhibited typical hydrostatic column behavior, with intermittent surface fluid discharges. Initial injection cycles yielded higher pressures and fluid volumes, while subsequent cycles showed a progressive decline in both parameters, indicating a reduction in displacement efficiency over time. These results provide valuable insights into the dynamic behavior of the system and will inform future optimization strategies.

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Nomenclature

N ₂	Nitrogen
BHT	Bottomhole Temperature
CT	Coiled Tubing
CTGL	Coiled Tubing Gas Lift
POOH	Pull Out Of Hole
RIH	Running In Hole
GS	Type of Pulling Tool
WPCE	Well Pressure Control Equipment

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